

NILUT Model Training — Todo Checklist

Step-by-step guide for retraining with new reference & content images

1. Current State (Before Changes)

Current Directory Structure:

```
test_images/training_data/
■■■ neon/input/           → 36 content images
■■■ portrait/input/       → 10 content images
■■■ reference/
  ■■■ natural_clean_film/  → ~11 reference images
  ■■■ moody_cinematic/    → ~12 reference images
  ■■■ bright_airy_cream_whites/ → ~11 reference images
  ■■■ high_contrast_street_editorial/ → ~12 reference images
  ■■■ warm_vintage_kodachrome/ → ~11 reference images
```

Item	Current Count
Content images (neon + portrait)	46
Reference images (5 styles)	~57
Per-style models	5 (.pt files)
Universal model	1 (universal.pt)
train.py ref image limit	First 6 per style ([:6])

2. Todo Checklist

Step 1: Add New Reference Images (40 per style)

Add 40 new reference images to each existing style folder:

Folder	Existing	Add	Total (Train)	Keep Aside (Test)
natural_clean_film/	~11	+40	~41	10
moody_cinematic/	~12	+40	~42	10
brightairy_cream_whites/	~11	+40	~41	10
high_contrast_street_editorial/	~12	+40	~42	10
warm_vintage_kodachrome/	~11	+40	~41	10

Total: ~200 new reference images for training + 50 for testing = 250 new images

Step 2: Create Test/Validation Folder

Create a separate folder for test images (10 per style). These should NOT be used in training.

```
test_images/validation/
  ■■■ natural_clean_film/      → 10 test images
  ■■■ moody_cinematic/        → 10 test images
  ■■■ brightairy_cream_whites/ → 10 test images
  ■■■ high_contrast_street_editorial/ → 10 test images
  ■■■ warm_vintage_kodachrome/ → 10 test images
```

Keep test images separate — never mix with training data. Use these to verify the model learned the style, not just memorized images.

Step 3: Add Landscape Content Images

Create a new content image directory with 20 landscape images:

```
test_images/training_data/
  ■■■ neon/input/             → 36 images (existing)
  ■■■ portrait/input/         → 10 images (existing)
  ■■■ landscape/input/        → 20 images (NEW)
  ■■■ reference/              → style folders
```

New total content images: 36 (neon) + 10 (portrait) + 20 (landscape) = 66 content images

Step 4: Update train.py — Reference Image Limit

Change the reference image limit from 6 to 50.

Current code (Line 63):

```
ref_images = sorted(glob.glob(f"{style_path}/*.jpg"))[:6]
```

Change to:

```
ref_images = sorted(glob.glob(f"{style_path}/*.jpg"))[:50]
```

Without this change, the training will only use the first 6 reference images per style and ignore all the new ones you added!

Step 5: Update train.py — Add Landscape Content Images

Update Line 46 to include the new landscape directory.

Current code (Line 46):

```
content = [cv2.imread(f) for f in
            glob.glob(f"{BASE}/neon/input/*.jpg") +
            glob.glob(f"{BASE}/portrait/input/*.jpg")]
```

Change to:

```
content = [cv2.imread(f) for f in
            glob.glob(f"{BASE}/neon/input/*.jpg") +
            glob.glob(f"{BASE}/portrait/input/*.jpg") +
            glob.glob(f"{BASE}/landscape/input/*.jpg")]
```

Step 6: Run Training

```
cd /home/shreyank06/Desktop/projects/upwork_projects/style_transfer
python train.py
```

- Choose **Option 1** to train all 5 per-style models + universal model
- Choose **Option 2** to train only the universal model (faster)

What happens during training:

Action	Details
1. Load content images	66 images (neon + portrait + landscape)
2. Per-style training	For each of 5 styles: stack up to 50 ref images, train MLP, save .pt
3. Backup old universal	Moves existing universal.pt to models/nilut/universal/{timestamp}/
4. Train universal model	Uses all references from all styles combined

Action	Details
5. Save new universal	Saves to models/nilut/latest/universal.pt
6. Update meta.json	Records timestamp, sample count, epochs

Step 7: Verify & Test

- Check that new .pt model files are generated in **models/nilut/**
- Check that old universal model is backed up in **models/nilut/universal/{timestamp}/**
- Test with validation images (from **test_images/validation/**)
- Run the web app and test style transfer on new images

```
uvicorn app:app --host 0.0.0.0 --port 8000
```

3. Model Storage Summary

Model	Location	Backup?
natural_clean_film.pt	models/nilut/	NO — overwritten directly
moody_cinematic.pt	models/nilut/	NO — overwritten directly
bright_airy_cream_whites.pt	models/nilut/	NO — overwritten directly
high_contrast_street_editorial.pt	models/nilut/	NO — overwritten directly
warm_vintage_kodachrome.pt	models/nilut/	NO — overwritten directly
universal.pt (new)	models/nilut/latest/	YES — old moved to timestamped folder
universal.pt (old backup)	models/nilut/universal/{timestamp}/	Timestamped by file modified date

Per-style models get overwritten with no backup. If you want to keep old per-style models, manually copy them before retraining.

4. Final Directory Structure (After All Changes)

```
test_images/
├── training_data/
│   ├── neon/input/           → 36 content images
│   ├── portrait/input/       → 10 content images
│   ├── landscape/input/      → 20 content images (NEW)
│   ├── reference/
│   │   ├── natural_clean_film/ → ~51 ref images (11 + 40)
│   │   ├── moody_cinematic/    → ~52 ref images (12 + 40)
│   │   ├── bright_airy_cream_whites/ → ~51 ref images (11 + 40)
│   │   ├── high_contrast_street_editorial/ → ~52 ref images
│   │   └── warm_vintage_kodachrome/ → ~51 ref images (11 + 40)
│   └── validation/           → (NEW)
│       ├── natural_clean_film/ → 10 test images
│       ├── moody_cinematic/    → 10 test images
│       ├── bright_airy_cream_whites/ → 10 test images
│       ├── high_contrast_street_editorial/ → 10 test images
│       └── warm_vintage_kodachrome/ → 10 test images
```

5. Before vs After Summary

	Before	After
Content images	46 (neon + portrait)	66 (+ 20 landscape)
Reference images (training)	~57 (5 styles)	~257 (+ 200 new)
Reference images (testing)	0	50 (10 per style)
train.py ref limit	[:6]	[:50]
train.py content dirs	neon + portrait	neon + portrait + landscape
Per-style models	5	5 (retrained)
Universal model	1	1 (retrained, old backed up)

Quick checklist: (1) Add 40 ref images per style (2) Create validation folder with 10 per style (3) Create landscape/input/ with 20 images (4) Change [:6] to [:50] in train.py line 63 (5) Add landscape glob to train.py line 46 (6) Run python train.py (7) Test results

6. How Training Actually Works (Explained)

The content/target images are **not just loaded passively**. They are actively used in training. Here's what actually happens:

Step	What Happens
1. Load content images	Load all 66 content images (neon + portrait + landscape)
2. Sample content colors	Extract 50,000 random colors (A,B channels) from each content image
3. Sample reference colors	Extract 50,000 random colors (A,B channels) from reference images
4. Pair colors	Use histogram matching to pair content colors with reference colors (red with red, blue with blue)
5. Train MLP	Train the neural network on these pairs: content color → reference color

Role of Each Image Type:

Image Type	Role in Training
Content images (66)	Provide the INPUT side — colors the model learns FROM
Reference images (~257)	Provide the OUTPUT side — colors the model learns TO

What the model learns:

"When I see this color from a content image, convert it to that color from the reference image"

Without content images, the model would have nothing to learn from.
Without reference images, the model would have nothing to learn to.

Visual Flow:

